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Yuchen Mao

Machine Learning Engineer | VLMs & Agentic Systems

Google Scholar: Yuchen Mao
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Github | Linkedin | Kaggle

Profile

Machine learning engineer with a strong base in fine-tuning, VLMs, agentic systems, and applying ML in specialist domains. Outside my day-to-day work, I compete on Kaggle to explore unfamiliar problems, learn quickly from real constraints, and bring those lessons back to work when building models and tools.

Skills

Languages	Python, Rust, R, SQL, LaTeX.
Tools	Git, Linux, Docker, Neo4j, Qdrant, Hugging Face, vLLM, SGLang, LangChain, LLM APIs.
ML Libraries	PyTorch, scikit-learn, pandas, SciPy, Plotly, XGBoost, LightGBM.
ML Methods	VLMs, LLM fine-tuning, LoRA, RLVR/GRPO, DPO, prompt optimization, agent evaluation, RAG.
Math	Statistics, probability, stochastic processes, time series analysis, probabilistic ML, Bayesian analysis.

Work Experience

Viridien | Machine Learning Engineer **2025/08 – Present**

- **VLM document classification**: Fine-tuned vision-language models for oil & gas technical documents using LoRA SFT, RLVR methods (GRPO/GSPO), and prompt optimization with ACE-Agent and DSPy. Improved F1 / accuracy from 0.47 / 0.74 to 0.78 / 0.85 at comparable model size.
- **Agentic label QC**: Built a QC system for noisy technical-document labels with domain experts. Converted expert review patterns into vision-enabled checks, domain "skills.md" knowledge, and a critic-validator loop to make label decisions easier to audit.
- **Agent terminal-use training**: Fine-tuned Qwen models on multi-GPU infrastructure with Harbor, SGLang, and AREAL to improve terminal-use behavior, in collaboration with CAMEL-AI.

Autodesk | Intern, Technology Consultant (AI/ML) **2024/06 – 2024/09**

- **VLM benchmarking for CAM**: Created a benchmark dataset for evaluating how vision-language models understand Computer-Aided Manufacturing workflows. The benchmark supported model selection for Fusion agents and gives the team a repeatable way to test new model releases.

Business-Intelligence of Oriental Nations Corporation Ltd | Intern, Algorithm Engineer **2023/06 – 2023/09**

- **Manufacturing ML**: Trained machine learning models for manufacturing applications using scikit-learn and XGBoost.

Selected Competitions & Hackathons

Kaggle Hackathon Honorable Mention (8/321) - [Solution Write-up] **May 2026**
Fine-tuned Gemma 2 2B Instruct for reasoning with a curated data mixture, LoRA reasoning SFT, and DPO based on the Delta Learning Hypothesis. Trained on Google TPU.

Kaggle Competition Master (Ranked 842 worldwide; 1x Gold, 2x Silver). **Feb 2026**

Kaggle Competition Gold Medal (13/3357) - [Solution Write-up] **Feb 2026**
Built a Rust solver for irregular 2D bin packing, placing non-convex shapes into the smallest possible bounding square across hundreds of configurations. Combined simulated annealing, local search, and constraint relaxation over the 3-month competition.

Kaggle Competition Silver Medal (19/1151) - Presented at NeurIPS 2025 Special Session [Solution Presentation] **Oct 2025**
Built a program-synthesis agent with an evolutionary database to solve ARC-AGI tasks while optimizing for minimal code length.

Mistral AI Award - Hack the Law Hackathon **Jun 2025**
Built a Mixtral-based multi-agent system to extract, classify, and link regulated activities and entities from legal documents.

1st Place - Infosys Cambridge AI Centre Hackathon **May 2025**
Built **Memory Plus**, an MCP memory layer for LLMs with a local vector store and RAG. (**50+ GitHub stars; 8k downloads**)

Publications

SeLoRA: Self-Expanding LoRA for High-Quality and Efficient Medical Image Synthesis **Feb 2026**
Expert Systems with Applications • Project Page

Semi-Supervised Medical Image Segmentation via Knowledge Mining from Large Models **Jan 2026**
arXiv Preprint • Accepted to IEEE International Symposium on Biomedical Imaging (ISBI)

Education

University of Cambridge, MPhil in Data Intensive Science (Distinction) **2024/10 - 2025/07**
Advanced study in statistics, machine learning, high-performance computing, and data-intensive scientific applications.

University of Edinburgh, BA in Mathematics (First Class Honours) **2020/09 - 2024/05**
Mathematics degree with emphasis on statistics, machine learning, and mathematical foundations for modelling.